Continuous, Categorical

Loan Status

Yes → Encoding

No

Yes

No

Yes

Yes

Encoding

Label Encoding

One Hot Encoding

Positional

Frequency Encoding

⋮

Flow of Name | generally use Label Encoding

Flow of Name	LEnco	Lily	Rose	Tulip
Lily	0	1	0	0
Rose	1	0	1	0
Lily	0	1	0	0
Rose	1	0	1	0
Tulip	2	0	0	1
Rose	1	0	1	0
Rose	1	0	1	0
Ross	1	0	1	0
Tulip	2	0	0	1

$2 > 1 > 0$

Tulip > Rose > Lily

Continuous Data Transformations

Data transformation

- Linear
- Non-Linear Transformation

Salary	Age	work
1000000	21.2	1
5000000	22.5	2
7000000	23.4	3

-1 to 1   -1 to 1   1 to 1

Outliers

data transformation

$$\xrightarrow{\text{Linear Transformation}} \quad y = ax + b$$

Normalization

Standardization

$$\frac{x_i - X_{min}}{X_{max} - X_{min}}$$

$$\frac{x_i - \mu}{\sigma}$$

Normalization — $\frac{x_i - x_{min}}{x_{max} - x_{min}}$

standardization — $\frac{x_i - \mu}{\sigma}$

x

7 $\rightarrow \frac{7-1}{8-1} \Rightarrow \frac{6}{7} \Rightarrow 0.85$

8 $\rightarrow \frac{8-1}{8-1} \Rightarrow \frac{7}{7} \Rightarrow 1$

1 $\rightarrow \frac{1-1}{8-1} \Rightarrow 0$

2 $\rightarrow \frac{2-1}{8-1} \Rightarrow \frac{1}{7} \Rightarrow 0.14$

3 $\rightarrow \frac{3-1}{7} \Rightarrow \frac{2}{7} \Rightarrow 0.28$

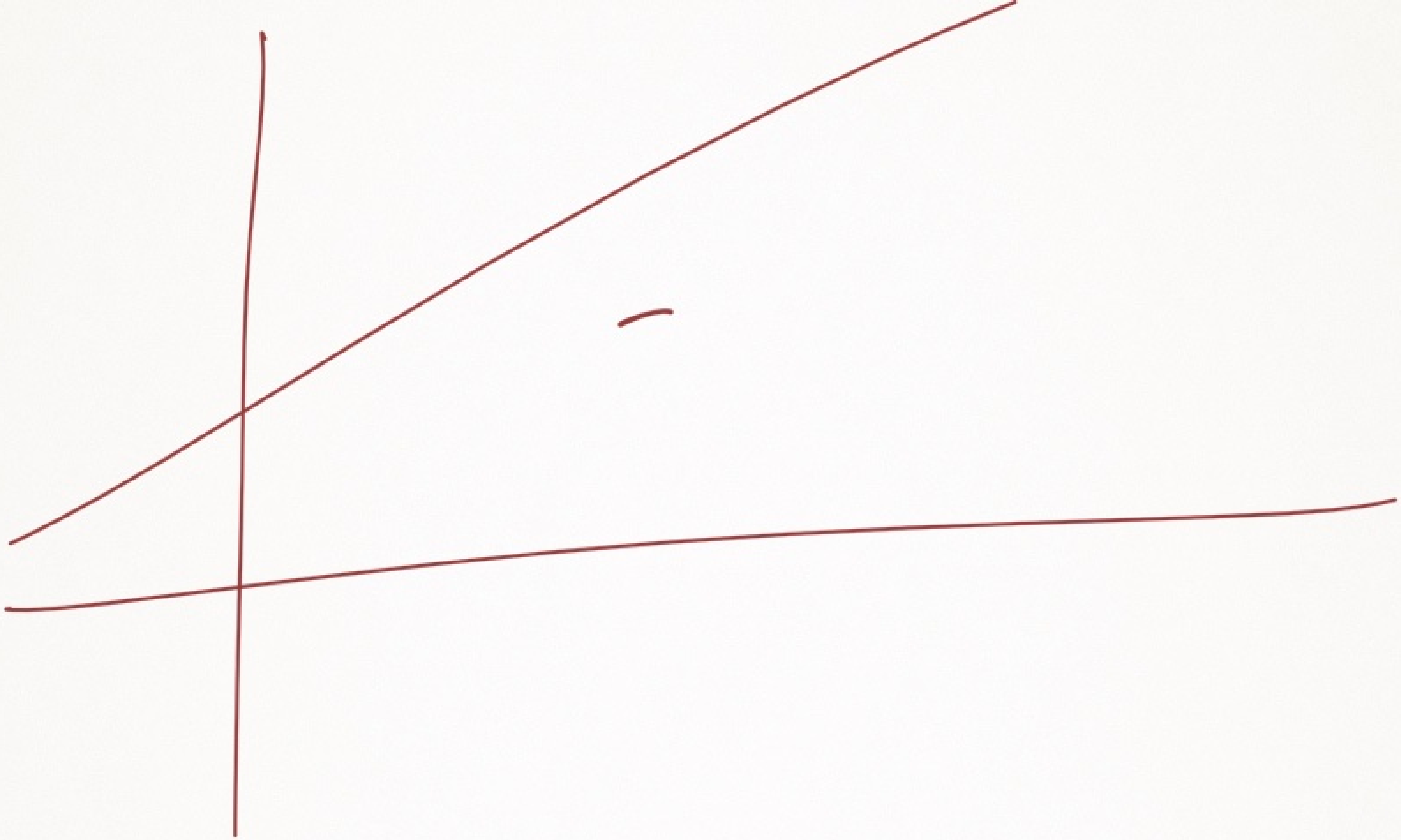
Non-Linear Traformation

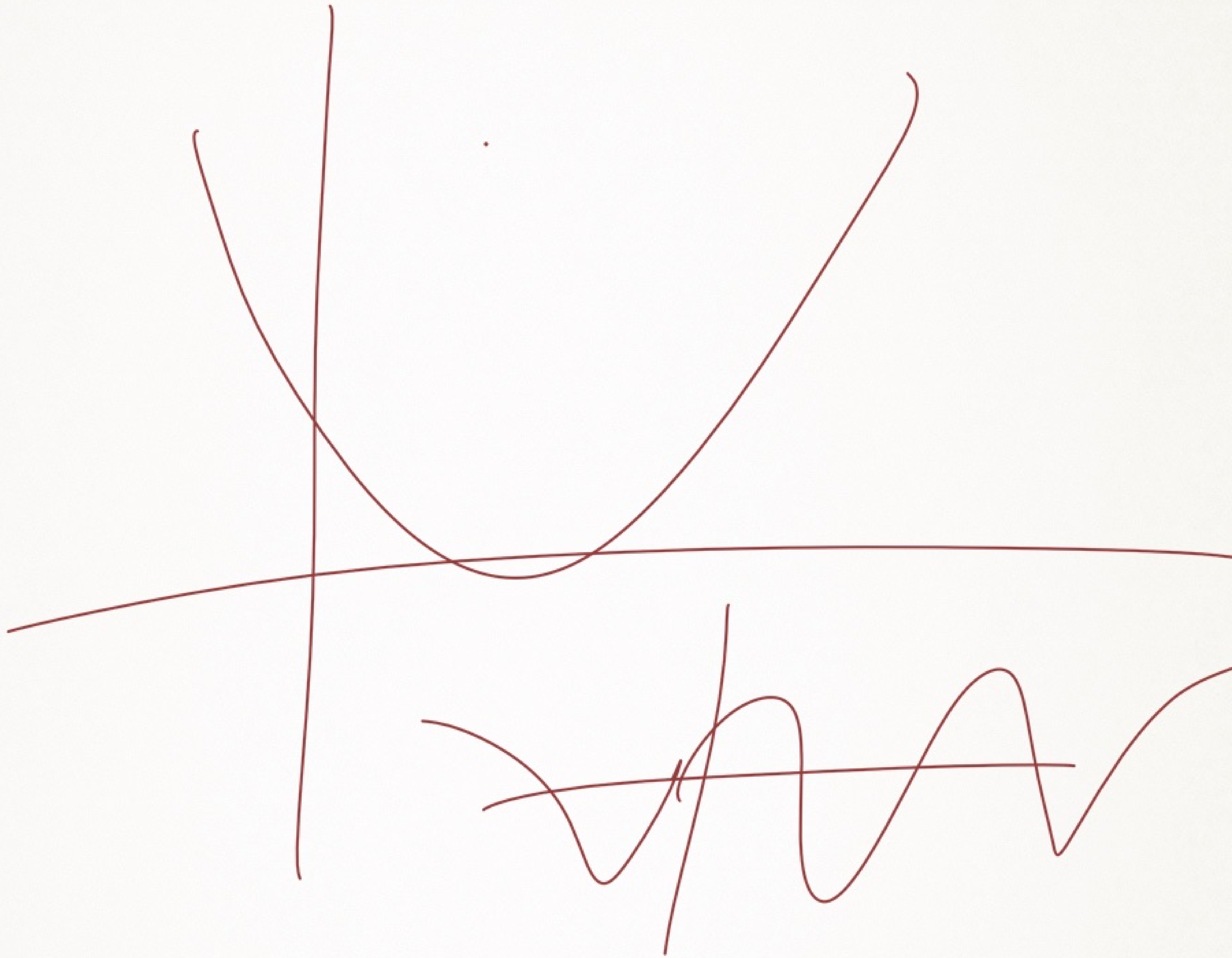
$$\frac{x_i - \mu \leftarrow \text{mean}}{\sigma \leftarrow \text{std}}$$

$$\left(\frac{1}{\sigma}\right) x_i - \left(\frac{\mu}{\sigma}\right)$$

$$\left(\frac{1}{\sigma}\right) x_i + \left(\frac{-\mu}{\sigma}\right)$$

$$\boxed{a \quad x \quad + \quad b}$$





Non-lin

→ Power transformation $\Rightarrow x^2, x^3$

$x^{-1/2}$

→ $\log x$

e^x

Preprocessing → model Selection

split (train test ~~Validation~~)

100% data → model

70	30
75	25
80	20
85	15

train $\rightarrow$ model
fit train

testing $\rightarrow$ Evaluation

Result [illegible]

hyperparameter

[illegible] [illegible]

train test

train model

Evaluate